P1: ENF and continuous closest-point correspondence

Actual finite STSP data, deterministic population quadrature, and a constructed counterexample

A. Finite STSP: medians over ten original parents, 32 exact shared references each

Method	E_∞	E_{avg}	E_{match}	E_{base}	Total CPU (s)	Failed queries
Oracle	0.0227381	0.0100548	0	1.54937e-14	2.09676	0/320
ENF	0.022716	0.0100516	0.00138141	0.00138572	2.27625	0/320
CP	0.0227369	0.0100537	0.000171198	0.000171599	25.5622	0/320
NN (32)	0.0418692	0.0209652	0.0968374	0.0968377	2.18484	0/320
NN (128)	0.0218451	0.0102646	0.031412	0.0314085	6.56621	0/320
NN (512)	0.0227292	0.0100538	0.00130896	0.0013087	22.731	0/320

Timing includes empirical target representation and search. It is implementation-specific.

B. Population matching accuracy: seven noise levels, no Monte Carlo

Method	Slope, all seven	Slope, smallest four	$E_{\text{match}}/\sigma^4$ range
ENF	4.002706	4.001245	0.599781 to 0.602043
CP	4.001560	4.000724	1.32416 to 1.32703

Finite-range slopes and resolution checks are numerical evidence, not an asymptotic proof.

C. Oscillatory normal graph: local ENF uniqueness versus nonunique global CP

N = 15625; a = 0.00000256; reference radius = 1.01; target mean radius = 1.04.

Two CP angles: +/-0.000190347562617 rad; local ENF point: (1.04000256, 0).

g(0) = 0.0009001536065536; g(CP) = 0.000899886612010056.

Limiting CP branch jump = 0.000395921966899; ENF varies continuously.

The graph is constructed, not STSP-generated. Nonuniqueness does not disprove fourth-order position accuracy.

Continuous CP in the empirical test searches the verified local fitted sheet; no theorem certifies all spurious components.